

## **SECTION 01 6116 - VOLATILE ORGANIC COMPOUND (VOC) CONTENT RESTRICTIONS**

### **PART 1 GENERAL**

#### **1.01 SECTION INCLUDES**

- A. VOC restrictions for product categories listed below under "DEFINITIONS."
- B. All products of each category that are installed in the project must comply; the Government's project goals do not allow for partial compliance.

#### **1.02 RELATED REQUIREMENTS**

- A. Section 01 3001 - Submittals: Submittal procedures.
- B. Section 01 4000 - Quality Requirements: Procedures for testing and certifications.
- C. Section 01 5719 - Temporary Environmental Controls: Procedures and testing.
- D. Section 01 6000 - Product Requirements: Fundamental product requirements, substitutions and product options, delivery, storage, and handling.

#### **1.03 DEFINITIONS**

- A. VOC-Restricted Products: All products of each of the following categories when installed or applied on-site in the building interior:
  - 1. Adhesives, sealants, and sealer coatings.
  - 2. Carpet.
  - 3. Carpet cushion.
  - 4. Carpet tile.
  - 5. Resilient floor coverings.
  - 6. Wood flooring.
  - 7. Paints and coatings.
  - 8. Insulation.
  - 9. Gypsum board.
  - 10. Acoustical ceilings and panels.

11. Cabinet work.
12. Wall coverings.
13. Composite wood and agrifiber products used either alone or as part of another product.
14. Other products when specifically stated in the specifications.

F-Interior of Building: Anywhere inside the exterior weather barrier.

G-Adhesives: All gunnable, trowelable, liquid-applied, and aerosol adhesives, whether specified or not; including flooring adhesives, resilient base adhesives, and pipe jointing adhesives.

H-Sealants: All gunnable, trowelable, and liquid-applied joint sealants and sealant primers, whether specified or not; including firestopping sealants and duct joint sealers.

#### **1.04 REFERENCE STANDARDS**

- A. CAL (CHPS LEM) - Low-Emitting Materials Product List; California Collaborative for High Performance Schools (CHPS); current edition at [www.chps.net/](http://www.chps.net/).
- B. CAL (CDPH SM) - Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions From Indoor Sources Using Environmental Chambers; California Department of Public Health v1.1, 2010.
- C. CRI (GLCC) - Green Label Testing Program - Approved Product Categories for Carpet Cushion; Carpet and Rug Institute; Current Edition.
- D. CRI (GLP) - Green Label Plus Testing Program - Certified Products; [www.carpet-rug.org](http://www.carpet-rug.org) current edition.
- E. UL (GGG) - GREENGUARD Gold Certified Products; UL Environment; current listings at <http://productguide.ulenvironment.com/QuickSearch.aspx>.
- F. GreenSeal GS-36 - Commercial Adhesives 2011.
- G. SCAQMD 1168 - South Coast Air Quality Management District Rule No.1168 current edition.
- H. SCS (CPD) - SCS Certified Products current listings at [www.scs-certified.com](http://www.scs-certified.com).

#### **1.05 SUBMITTALS**

- A. See Section 01 3000 - Administrative Requirements, for submittal procedures.

- B. Evidence of Compliance: Submit for each different product in each applicable category.
- C. Product Data: For each VOC-restricted product used in the project, submit evidence of compliance.

## **PART 2 PRODUCTS**

### **2.01 MATERIALS**

- A. All VOC-Restricted Products: Provide products having VOC content of types and volume not greater than those specified in State of California Department of Health Services Standard Practice for the Testing of Volatile Organic Emissions From Various Sources Using Small-Scale Environmental Chambers.
  - 1. Evidence of Compliance: Acceptable types of evidence are:
    - a. Current GREENGUARD Children & Schools certification; [www.greenguard.org](http://www.greenguard.org).
    - b. Current Carpet and Rug Institute Green Label Plus certification; [www.carpet-rug.org](http://www.carpet-rug.org).
    - c. Current SCS Floorscore certification; [www.scs-certified.com](http://www.scs-certified.com).
    - d. Current SCS Indoor Advantage Gold certification; [www.scs-certified.com](http://www.scs-certified.com).
    - e. Product listing in the CHPS Low-Emitting Materials Product List at [www.chps.net/manual/lem\\_table.htm](http://www.chps.net/manual/lem_table.htm).
    - f. Current certification by any other agencies acceptable to CHPS.
    - g. Report of laboratory testing performed in accordance with CHPS requirements for getting a product listed in the Low-Emitting Materials Product List; report must include laboratory's statement that the product meets the specified criteria.
- B. Adhesives and Joint Sealants: Provide only products having volatile organic compound (VOC) content not greater than required by South Coast Air Quality Management District Rule No.1168.
  - 1. Evidence of Compliance: Acceptable types of evidence are:
    - a. Report of laboratory testing performed in accordance with requirements.
    - b. Published product data showing compliance with requirements.

- c. Certification by manufacturer that product complies with requirements.

F-Aerosol Adhesives: Provide only products having volatile organic compound (VOC) content not greater than required by GreenSeal GS-36.

1. Evidence of Compliance: Acceptable types of evidence are:
  - a. Current GreenSeal Certification.
  - b. Report of laboratory testing performed in accordance with GreenSeal GS-36 requirements.
  - c. Published product data showing compliance with requirements.

G-Paints and Coatings:

1. Provide coatings that comply with the most stringent requirements specified in the following:
  - a. 40 CFR 59, Subpart D--National Volatile Organic Compound Emission Standards for Architectural Coatings.
  - b. Ozone Transport Commission (OTC) Model Rule, Architectural, Industrial, and Maintenance Coatings; [www.otcair.org](http://www.otcair.org); specifically:
2. Determination of VOC Content: Testing and calculation in accordance with 40 CFR 59, Subpart D (EPA Method 24), exclusive of colorants added to a tint base and water added at project site; or other method acceptable to authorities having jurisdiction.
3. Evidence of Compliance: Acceptable types of evidence are:
  - a. Report of laboratory testing performed in accordance with requirements.
  - b. Published product data showing compliance with requirements.
  - c. Certification by manufacturer that product complies with requirements.

H-Carpet and Adhesive: Provide products having VOC content not greater than that required for CRI Green Label Plus certification.

1. Evidence of Compliance: Acceptable types of evidence are:
  - a. Current Green Label Plus Certification.
  - b. Report of laboratory testing performed in accordance with requirements.

- F. Carpet Cushion: Provide products having VOC content not greater than that required for CRI Green Label certification.
  - 1. Evidence of Compliance: Acceptable types of evidence are:
    - a. Current Green Label Certification.
    - b. Report of laboratory testing performed in accordance with requirements.
- G. Carpet Tile and Adhesive: Provide products having VOC content not greater than that required for CRI Green Label Plus certification.
  - 1. Evidence of Compliance: Acceptable types of evidence are:
    - a. Current Green Label Plus Certification.
    - b. Report of laboratory testing performed in accordance with requirements.
- H. Composite Wood and Agrifiber Products and Adhesives Used for Laminating Them: Provide products having no added urea-formaldehyde resins.
  - 1. Evidence of Compliance: Acceptable types of evidence are:
    - a. Published product data showing compliance with requirements.

## **PART 3 EXECUTION**

### **3.01 FIELD QUALITY CONTROL**

- A. The Government reserves the right to reject non-compliant products, whether installed or not, and require their removal and replacement with compliant products at no extra cost to the Government.
- B. Additional costs to restore indoor air quality due to installation of non-compliant products will be borne by Contractor.

### **END OF SECTION**